Maldigestion and Malabsorption:

Manifestations – Mediators – Mitigators

Nicole Meier, ND

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Disclosures

Nicole Meier, ND, disclosed no relevant financial relationships with any ineligible companies.

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Learning Objectives

At the conclusion of this segment, successful participants will be able to:

1. List the five dysfunctions of the DIGIN mnemonic.

2. Contrast the normal physiology of digestion and absorption with the pathological mechanisms of maldigestion and malabsorption.

3. List common clinical findings, antecedents, triggers, and mediators of maldigestion and malabsorption.

4. Explain how maldigestion and malabsorption may serve as mediators of extraintestinal disease and/or dysfunction.

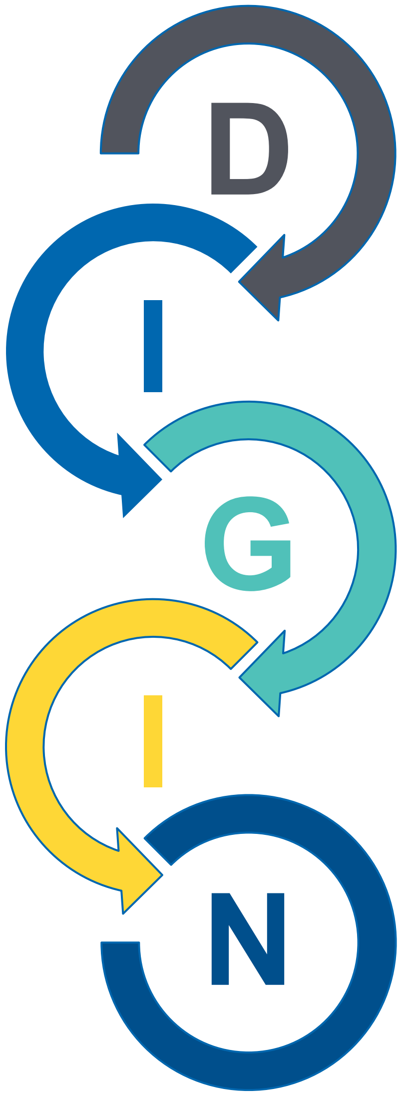
IFM Clinical Skills and Performance Objectives

THIS SEGMENT WILL ENABLE PARTICIPANTS TO DEVELOP & APPLY THESE SKILLS IN THEIR PRACTICE:

1. Identify and document the patient's clinical findings of maldigestion and malabsorption and elicit the antecedents, triggers, and mediators thereof on IFM's Functional Medicine Timeline and Matrix.

2. Identify extraintestinal diseases and dysfunctions that may be mediated by maldigestion and malabsorption.

The DIGIN Mnemonic of Gut Dysfunctions



***D**igestion/Absorption*

***I**ntestinal Permeability*

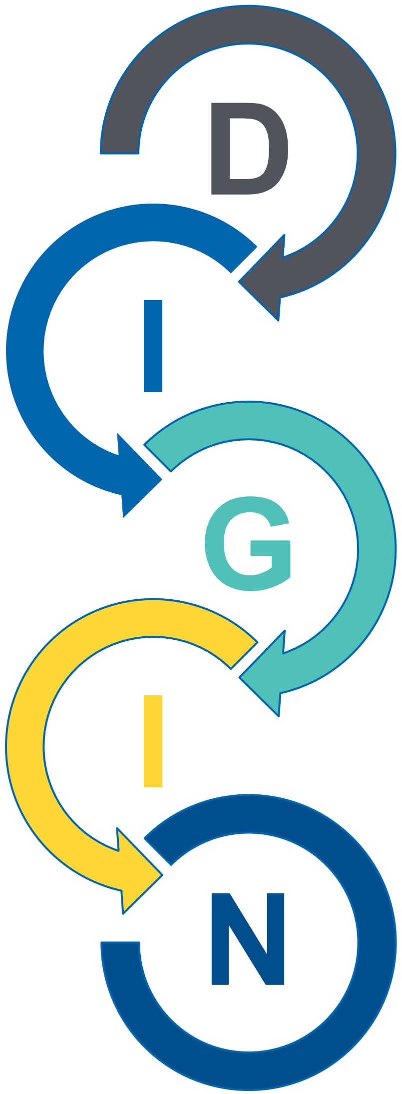
***G**ut Microbiota/Dysbiosis*

***I**nflammation/Immune*

***N**ervous System*

The DIGIN Model was developed collaboratively by Liz Lipski, PhD, CNS, Patrick Hanaway, MD, and IFM

The DIGIN Mnemonic of Gut Dysfunctions



***D**igestion/Absorption*

The DIGIN Model was developed collaboratively by Liz Lipski, PhD, CNS Patrick Hanaway, MD, and IFM

Clinical Focus

- Normal physiology of digestion and absorption

Area of Contact With the Outside World

GI mucosal surface area:
30-40 m^2

Begins in the oral cavity and ends at the anus

Helander HF, Fändriks L. Surface area of the digestive tract - revisited. *Scand J Gastroenterol*. 2014;49(6):681-689. doi:10.3109/00365521.2014.898326

Phases of Digestion

Cephalic phase

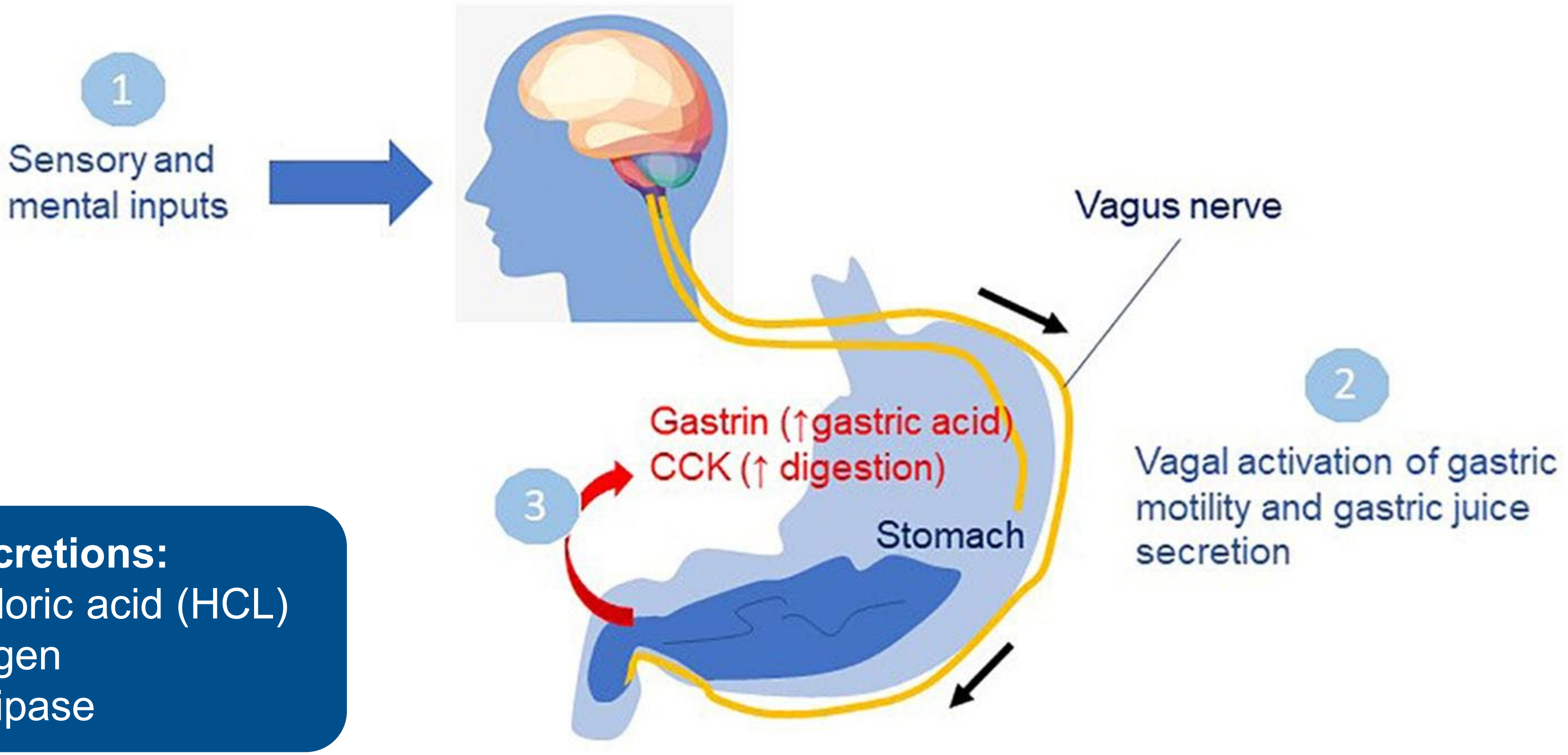
Gastric phase

Intestinal phase

Digestion: A Symphony of Harmony

Phases of Digestion: Cephalic Phase

Salivary secretions:
- Amylase
- Lipase
- Mucin



Gastric secretions:
- Hydrochloric acid (HCL)
- Pepsinogen
- Gastric lipase

- Fine LG, Riera CE. Sense of smell as the central driver of Pavlovian appetite behavior in mammals. *Front Physiol*. 2019;10:1151. doi:10.3389/fphys.2019.01151

Long Description Phases of Digestion: Cephalic Phase

- Sensory and mental inputs activate the cephalic phase of digestion via the vagus nerve, prompting the release of salivary amylase, lipase, and mucin to initiate digestion in the mouth. Vagal activation also stimulates gastric motility and gastric juice secretions, including hydrochloric acid, pepsinogen, and gastric lipase. C C K also begins to rise.

Mechanical and Chemical Digestion
Clinical Takeaway: Chew Thoroughly; Eat Mindfully

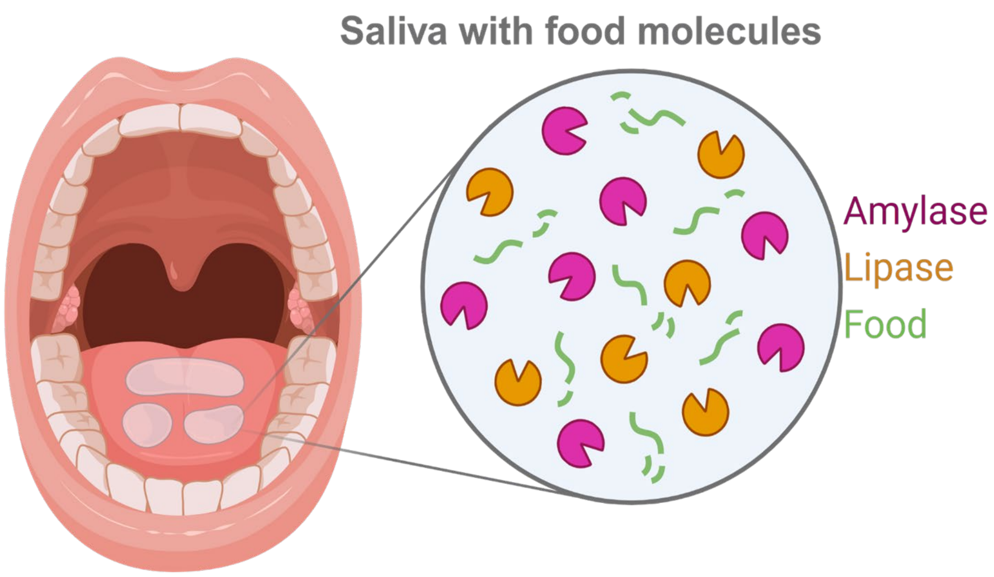


Image: Adapted from "You Can't Taste Food Without Saliva," by BioRender.com (2024) Retrieved from: https://app.biorender.com/biorender-templates/figures/all/t-6244990e5c8baf57fd918d4d-you-cant-taste-food-without-saliva by The Institute for Functional Medicine."

- Mechanical digestion
  - Grinding food into smaller pieces
  - Pushes food toward esophagus
- Chemical digestion
  - Salivary amylase
    - Breaks down starches/complex carbohydrates into simple sugars/disaccharides
  - Lingual lipase
    - Breaks down triglycerides into free fatty acids

Patricia JJ, Dhamoon AS. *Physiology, Digestion*. StatPearls Publishing; 2022. Updated September 12, 2022. Accessed June 2, 2025. https://www.ncbi.nlm.nih.gov/books/NBK544242/

Mechanisms of Chewing

- Proper mastication before swallowing food can affect the satiety response from hormones such as insulin and glucagon-like-peptide-1 (GLP1).

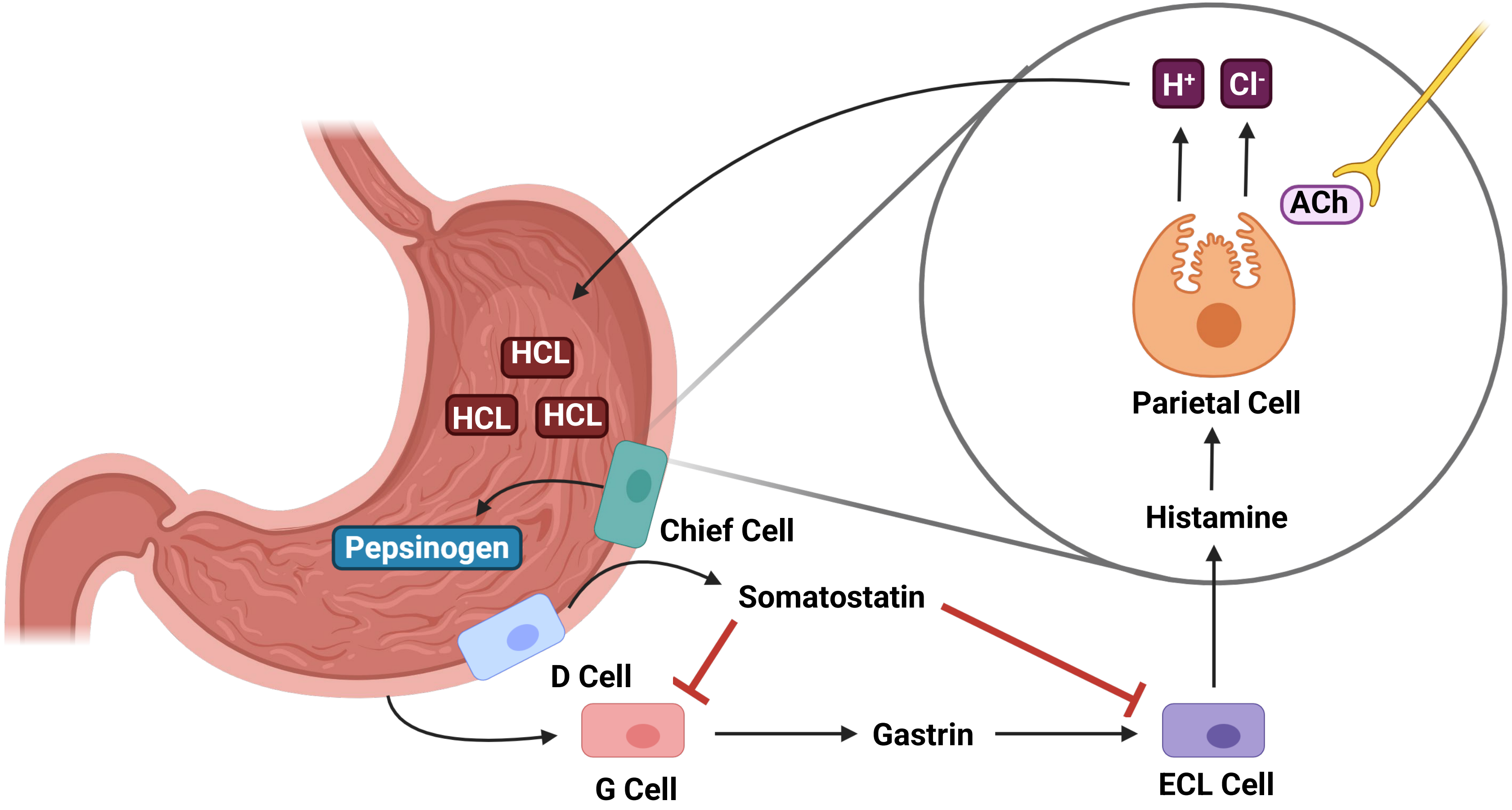
- In a study, it was concluded that chewing 30 times per bite increased the GLP-1 response in healthy volunteers.

Conclusions: Adequate chewing can increase the release of GLP-1, leading to feelings of fullness and managing blood sugar by prompting digestion.

- Sonoki K, Iwase M, Takata Y, et al. Effects of thirty-times chewing per bite on secretion of glucagon-like peptide-1 in healthy volunteers and type 2 diabetic patients. *Endocr J.* 2013;60(3):311-319. doi:10.1507/endocrj.ej12-0310
- Sardar A, Shah I, Malik O, Mahmood R, Asmatullah A, Qaisar O. Effects of chewing time of high protein meal on satiety and GLP-1 hormone. *Pak J Physiol.* 2024;20(2):31-35. doi:10.69656/pjp.v20i2.1632

Phases of Digestion: Gastric Phase

Clinical Takeaway: Improve Vagal/Parasympathetic Tone



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Long Description Phases of Digestion: Gastric phase

- In the gastric phase of digestion, mechanical and chemical signals cause G cells to release gastrin, stimulating the release of histamine from enterochromaffin-like cells, and promoting H C L secretion from parietal cells. Chief cells secrete pepsinogen, while D cells release somatostatin to inhibit G cells and E C L cells. Vagal tone modulates acetylcholine release, thereby coordinating digestion throughout this phase.

Phases of Digestion: Gastric Phase

- **Mechanical stimuli:** Entry of food into the stomach causes distention, which stimulates gastric activity through the vasovagal and local ENS reflexes (release ACh)

- **Chemical stimuli:** Partially digested proteins and amino acids stimulate G cells to release gastrin

Gastric secretions:
- Hydrochloric acid (HCL)
- Pepsinogen
- Gastric lipase
- Intrinsic factor
- Mucus

Gastric Enzymes

- Hydrochloric acid (HCL): Secreted by the parietal cells in the stomach
  - Creates acidic environment unsuitable for pathogens (optimal <pH 3)
  - Activates enzymes, including pepsinogen to the active form, pepsin

- Pepsinogen: Converts to active form, pepsin at low pH
  - Breaks down protein when gastric pH=2-3

- Intrinsic factor (IF): Protein produced by parietal cells in the stomach
  - Binds to vitamin B12 in the stomach
  - Essential for the absorption of B12 in the small intestine (ileum)

Patricia JJ, Dhamoon AS. *Physiology, Digestion*. StatPearls Publishing; 2022. Updated September 12, 2022. Accessed June 2, 2025. https://www.ncbi.nlm.nih.gov/books/NBK544242/

Phases of Digestion: Intestinal Phase

- **Mechanical digestion:** the mixing and movement of food, mainly though segmentation
- **Chemical digestion**: breaks down carbohydrates, fats, proteins and nucleic acids
  - Agents: amylase, lipase, peptidases
- **Nutrients absorbed:** peptides, amino acids, fructose, fats, water, minerals & vitamins

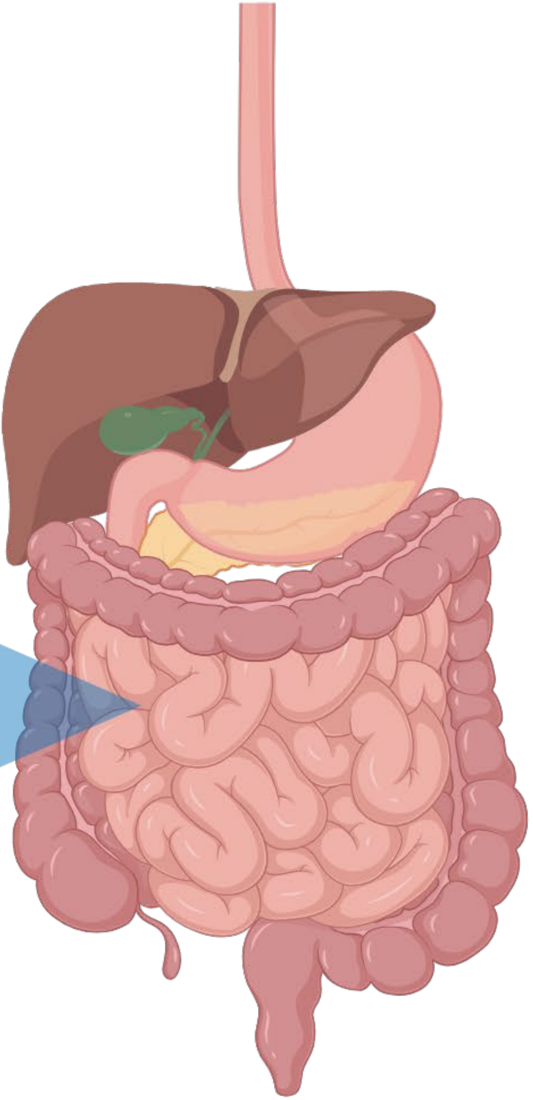


Image: Created with BioRender.com by The Institute for Functional Medicine

Long Description

- Mechanical digestion: the mixing and movement of food, mainly through segmentation
- Chemical digestion breaks down carbohydrates, fats, proteins, and nucleic acids
  - Agents: amylase, lipase, peptidases
- Nutrients absorbed: peptides, amino acids, fructose, fats, water, minerals, and vitamins

Phases of Digestion: Intestinal Phase

Within the Small Intestine:

- Chemical, not mechanical, is the main form of digestion.
- Cholecystokinin (CCK): Hormone released when fat and protein reach the small intestine, triggering contraction of gallbladder (bile) and pancreatic digestive enzymes.
- Digestive enzymes are also released from the brush border (microvilli) of enterocytes.

Patricia JJ, Dhamoon AS. *Physiology, Digestion*. StatPearls Publishing; 2022. Updated September 12, 2022. Accessed June 2, 2025. https://www.ncbi.nlm.nih.gov/books/NBK544242/

Brush Border Enzymes

- Disaccharidases: Degrade disaccharides into monosaccharides
  - Sucrase: Sucrose into glucose & fructose
  - Lactase: Lactose into glucose and galactose
  - Maltase: Maltose into 2 glucose
  - Isomaltase (alpha-dextrinase): Isomaltose into glucose, maltose, and oligosaccharides

- Proteases: Breaks down proteins into peptides and amino acids
  - Aminopeptidase
  - Carboxypeptidase
  - Dipeptidase

- Glucoamylase: Breaks down polysaccharides into glucose molecules

Kiela PR, Ghishan FK. Physiology of intestinal absorption and secretion. *Best Pract Res Clin Gastroenterol*. 2016;30(2):145-159. doi:10.1016/j.bpg.2016.02.007

Pancreatic Enzymes

- Amylase
- Lipase

Identical to salivary enzymes

- Peptidases: Break down proteins into smaller peptides
  - Mechanism: “Pro-enzymes” converted to active form:
    - Trypsinogen: Precursor to trypsin, activated in the small intestine (duodenum) by enzymes in the brush border
    - Chymotrypsinogen: Precursor to chymotrypsin, activated by trypsin
    - Procarboxypeptidase: Precursor to carboxypeptidase, activated by trypsin
    - Proelastase: Precursor to enzyme elastase, activated by trypsin

Patricia JJ, Dhamoon AS. *Physiology, Digestion*. StatPearls Publishing; 2022. Updated September 12, 2022. Accessed June 2, 2025. https://www.ncbi.nlm.nih.gov/books/NBK544242/

Bile

- Produced by hepatocytes and stored in the gallbladder
  - 95% of bile acids are reabsorbed in the distal small intestine
  - Bile acids are reused 20 times
- Primary mechanisms and functions:
  - Fat digestion via emulsification
  - Fat absorption via micelle formation
  - Excretion of toxins, toxicants, and waste products
  - Neutralization of stomach acid

Staley C, Weingarden AR, Khoruts A, Sadowsky MJ. Interaction of gut microbiota with bile acid metabolism and its influence on disease states. *Appl Microbiol Biotechnol*. 2017;101(1):47-64. doi:10.1007/s00253-016-8006-6

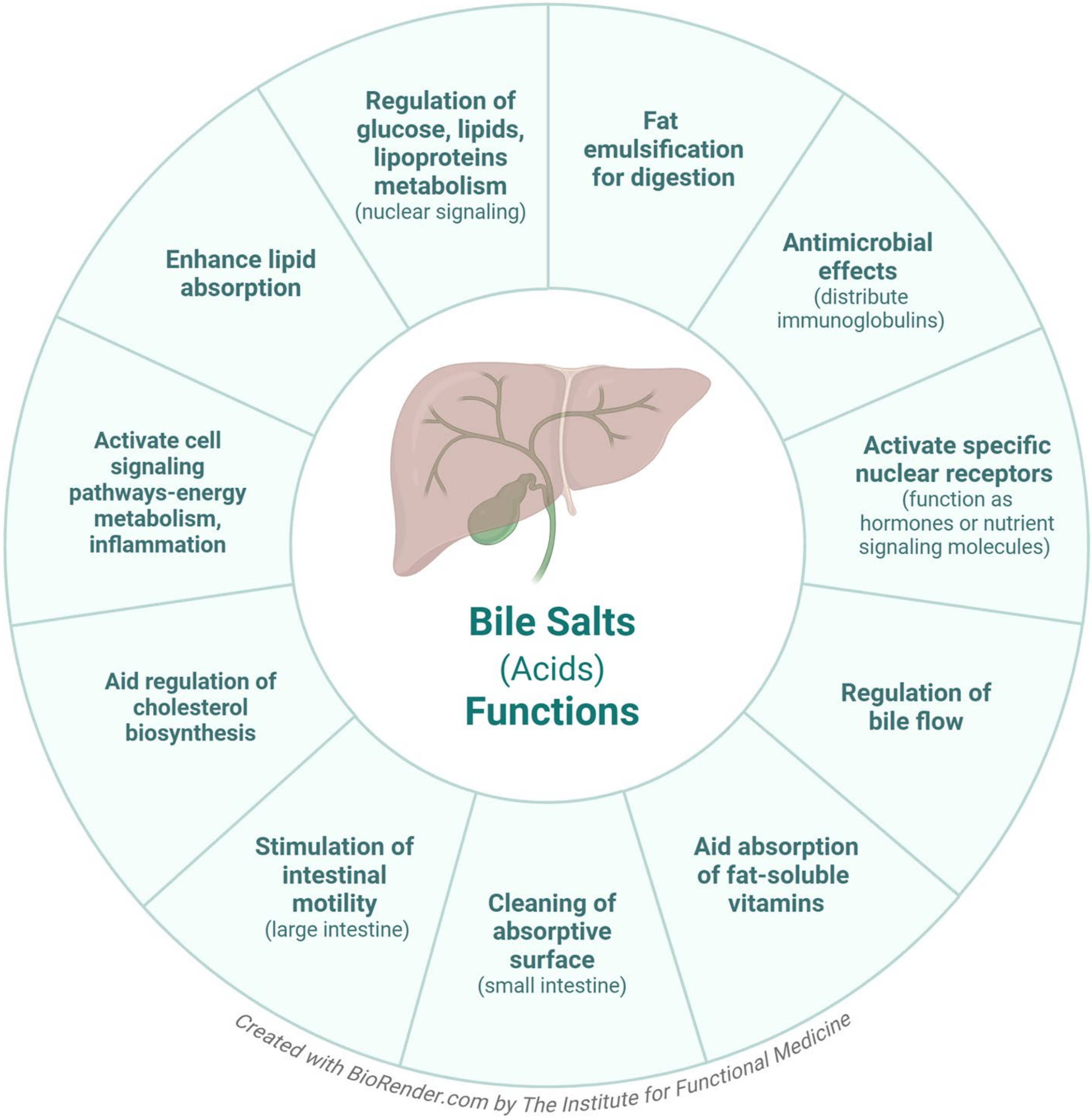
Bile Salts: Functions Beyond Digestion

- Antimicrobial effects
- Regulation of bile flow
- Signaling molecule for hormones and nutrients
- Regulation of glucose, lipids, lipoproteins, energy metabolism, and inflammatory response

Zhou H, Hylemon PB. Bile acids are nutrient signaling hormones. *Steroids*. 2014;86:62-68. doi:10.1016/j.steroids.2014.04.016

Bile Salts…

More Than Detergent Molecules



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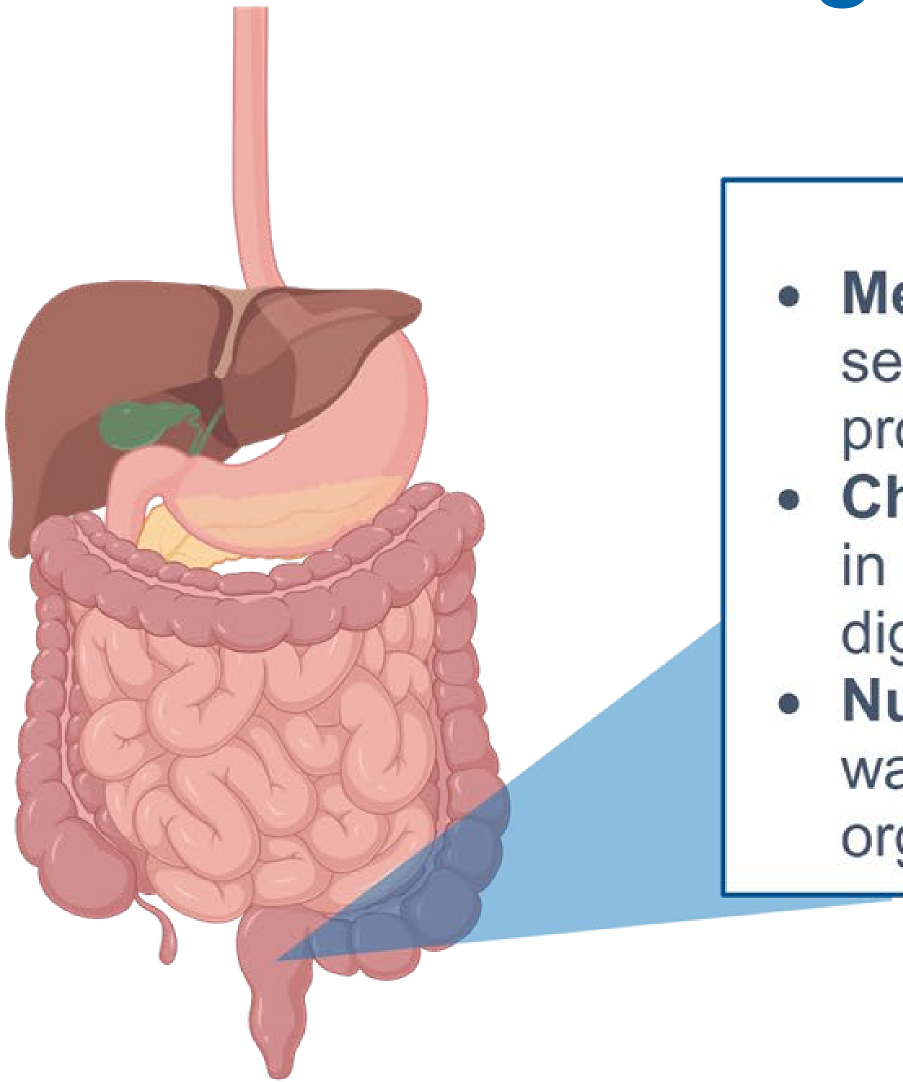
- Zhou H, Hylemon PB. Bile acids are nutrient signaling hormones. *Steroids*. 2014;86:62-68. doi:10.1016/j.steroids.2014.04.016

Long Description: Bile Salts Functions

- Bile salt functions
  - Regulation of glucose, lipids, lipoprotein metabolism through nuclear signaling
  - Fat emulsification for digestion
  - Antimicrobial effects through distribution of immunoglobulins
  - Activate specific nuclear receptors, function as hormones or nutrient signaling molecules
  - Regulation of bile flow
  - Aid absorption of fat-soluble vitamins
  - Cleaning of absorptive surface in small intestine
  - Stimulation of intestinal motility in large intestine
  - Aid regulation of cholesterol biosynthesis
  - Activate cell signaling pathways, energy metabolism, inflammation
  - Enhance lipid absorption

Intestinal Phase of Digestion: Colon



- **Mechanical digestion:** segmental mixing and propulsion via peristalsis
- **Chemical digestion:** bacteria in large intestine also perform digestive functions
- **Nutrients absorbed:** ions, water, minerals, vitamins and organic molecules

Image: Created with BioRender.com by The Institute for Functional Medicine

Long Description Intestinal phase of Digestion: Colon.

- Mechanical digestion
  - segmental mixing and propulsion via peristalsis
- Chemical digestion
  - Bacteria in large intestine also perform digestive functions
- Nutrients absorbed
  - Ions
  - Water
  - Minerals
  - Vitamins
  - Organic molecules

SUMMARY:

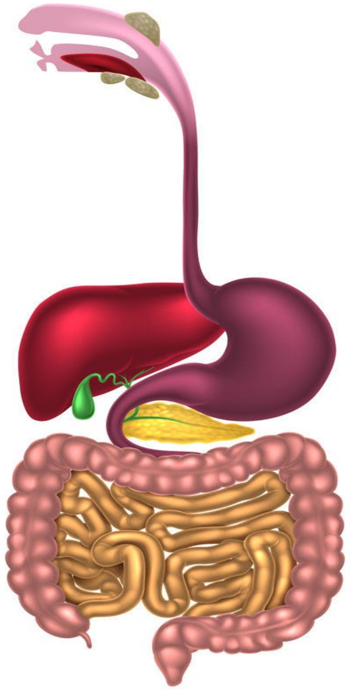
Intestinal Phase of Digestion by Location

Proximal Small Intestine (duodenum)	Fat MCT oil (absorbed throughout the GI tract) Sugars Peptides and amino acids Iron Folate Calcium (enhanced by vitamin D) Zinc Copper Water Electrolytes
Middle Small Intestine (jejunum)	Sugars Peptides and amino acids Calcium (enhanced by vitamin D) Magnesium Zinc Water Electrolytes
Distal Small Intestine (ileum)	Bile salts Water-soluble vitamins (B vitamins, vitamin C) Vitamin B_{12} (requires intrinsic factor) Fat-soluble vitamins (A,D,E,K) Magnesium Water Electrolytes
Colon	Water Electrolytes MCTs Amino acids

Kiela PR, Ghishan FK. Physiology of intestinal absorption and secretion. *Best Pract Res Clin Gastroenterol*. 2016;30(2):145-159. doi:10.1016/j.bpg.2016.02.007

Strive for Harmony

The Digestive Process



Organ	Movement	Agents	Macronutrients
Mouth	Chewing	Saliva	Starches
Esophagus	Peristalsis	None	None
Stomach	Upper muscle in stomach relaxes to let food enter and lower muscle mixes food with digestive juice	Stomach acid and digestive enzymes	Proteins
Small intestine	Peristalsis	Small intestine digestive enzymes	Starches, proteins, and carbohydrates
Pancreas	None	Pancreatic juice	Carbohydrates, fats, and proteins
Liver/gallbladder	None	Bile	Fat
Large intestine	Peristalsis	None	Bacteria in the large intestine can also break down food

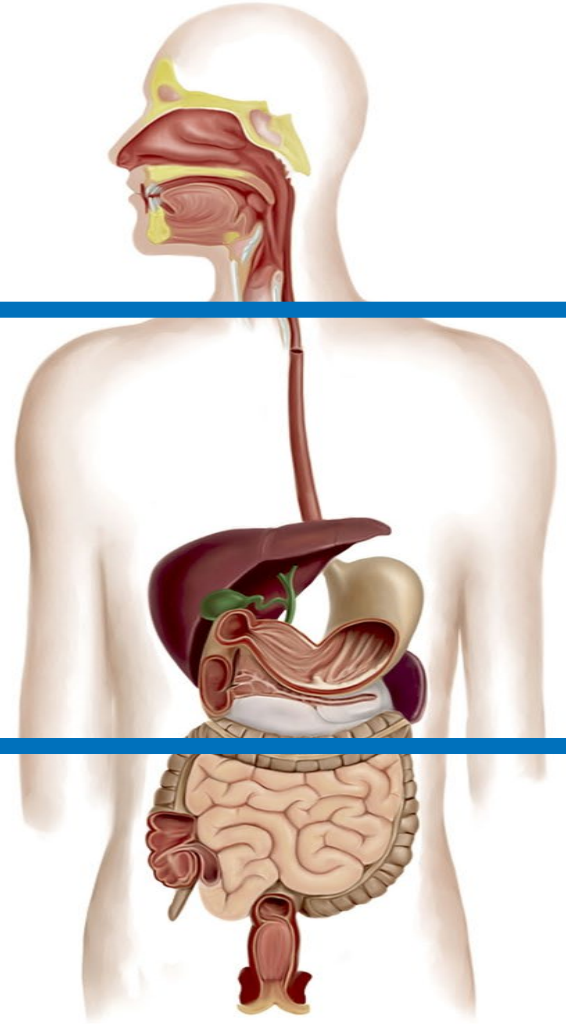
Based on: National Institute of Diabetes and Digestive and Kidney Diseases. Your digestive system & how it works. Reviewed December 2017. Accessed June 2, 2025. https://www.niddk.nih.gov/health-information/digestive-diseases/digestive-system-how-it-works

Clinical Focus

- Pathophysiological mechanisms of maldigestion and malabsorption.
- Antecedents, triggers, and mediators of maldigestion and malabsorption.

From Normal to Abnormal…

Normal Digestion → Maldigestion



Normal Absorption → Malabsorption

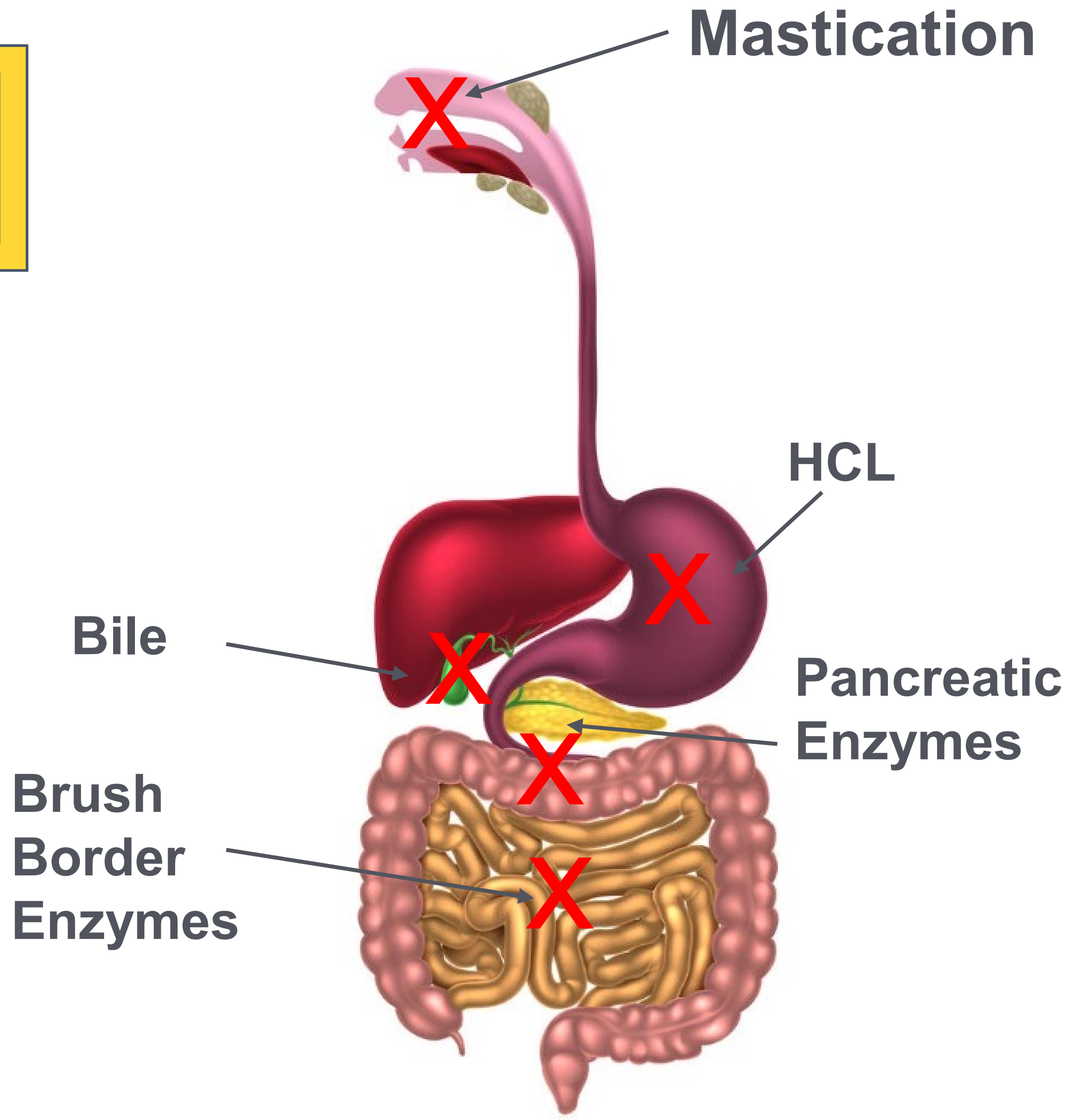
Terminology

The term *maldigestion* refers to defective hydrolysis of nutrients.

Whereas *malabsorption* refers to impaired mucosal absorption.

Impairments in Digestion and Absorption

- Inadequate mastication and salivary secretions
- Hypochlorhydria
- Gastric dysmotility
- Pancreatic insufficiency
- Bile insufficiency
- Brush border injury
- Intestinal dysmotility



Mechanisms, Mediators, and Manifestations of Maldigestion/Malabsorption: *Oral Factors*

Mechanism	Substrates/Nutrients Affected	Antecedents, Triggers, Mediators
Xerostomia (dry mouth)	**Carbohydrates, fats, proteins**	Medications, psychological factors (stress/anxiety), surgery, radiation, smoking, excessive coffee intake, dehydration, mouth breathing, older age
Inadequate chewing	**Carbohydrates, fats, proteins**	Dental issues, psychological factors (stress/anxiety), jaw issues (TMJ), decreased muscle control/tone

See: "References: Oral Factors"

References

ORAL FACTORS

Xerostomia:

- Tanasiewicz M, Hildebrandt T, Obersztyn I. Xerostomia of various etiologies: a review of the literature. *Adv Clin Exp Med*. 2016;25(1):199-206. doi:10.17219/acem/29375
- Müller F, Chebib N, Maniewicz S, Genton L. The impact of xerostomia on food choices—a review with clinical recommendations. *J Clin Med*. 2023;12(14):4592. doi:10.3390/jcm12144592

Medications as Mediators of Xerostomia

Many medications can cause xerostomia.

This list represents medications with an incidence of >10%:

Medication Class	Specific Medications
Anticholinergics	Atropine, belladonna, oxybutynin
Antidepressants	Citalopram, phenelzine
Antipsychotic	Haloperidol
Anxiolytics	Alprazolam, diazepam, triazolam
Anti-hypertensives	Captopril, lisinopril, enalapril
Diuretics	Furosemide, chlorothiazide, hydrochlorothiazide
Muscle relaxants	Tizanidine, cyclobenzaprine, orphenadrine
Analgesics	Opioids, NSAIDs
Antihistamines	Astemizole, loratadine, brompheniramine

Impairments in Digestion and Absorption

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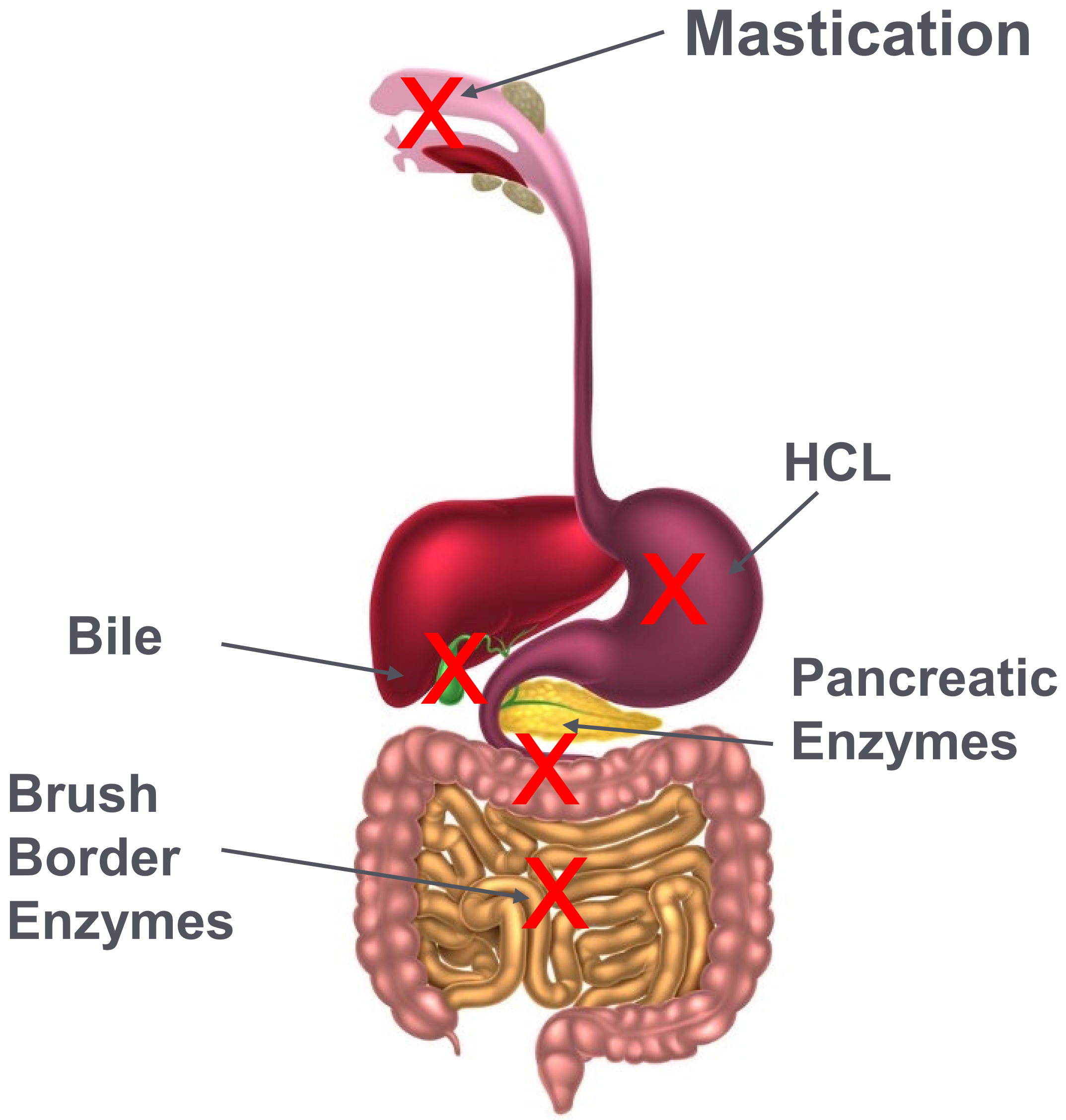

Mechanisms, Mediators, and Manifestations of Maldigestion/Malabsorption: *Gastric Factors*

Mechanism	Substrates/Nutrients Affected	Antecedents, Triggers, Mediators
Decreased gastric acid	**B12, folate, B6, iron, calcium, magnesium, protein**	Prior gastric resection, proton pump inhibitors, *H. pylori* infection, stress Atrophic gastritis, pernicious anemia, hypothyroidism, gastric cancer
Intrinsic factor insufficiency	**B12**	Prior gastric resection, proton pump inhibitors, alcohol consumption Atrophic gastritis, pernicious anemia
Rapid gastric emptying	**Fat, protein, carbohydrates**	Prior surgery Autonomic dysfunction

See: "References: Gastric Factors"

Mechanisms, Mediators, and Manifestations of Maldigestion/Malabsorption: *Gastric Factors*

Mechanism	Substrates/Nutrients Affected	Antecedents, Triggers, Mediators
Decreased gastric emptying (gastroparesis)	**Protein, fat, carbohydrates (hypoglycemia)**	Brainstem tumors, injury/trauma to brainstem, surgery, viral infections, medications, eating disorders Diabetes (most severe in type 1), rheumatological conditions (amyloidosis, scleroderma), autoimmune gastrointestinal dysmotility, neurologic conditions (Parkinson's disease, multiple sclerosis, autonomic neuropathy, dysautonomia)

**Prior gastric surgery (involving the pylorus and or including vagotomy) – accelerated emptying and decreased time for absorption in the small bowel; Gastroparesis – decreased gastric mixing and impaired protein and fat assimilation.*

See: "References: Gastric Factors"

References

GASTRIC FACTORS

Decreased Gastric Acid:

- Tan JT, Cheung CL, Cheung KS. Relationship between *Helicobacter pylori* infection, osteoporosis, and fracture. *J Gastroenterol Hepatol*. 2024;39(10):2006-2017. doi:10.1111/jgh.16664
- Guilliams TG, Drake LE. Meal-time supplementation with betaine HCl for functional hypochlorhydria: what is the evidence? *Integr Med (Encinitas)*. 2020;19(1):32-36.

Intrinsic Factor Insufficiency:

- Al-Awami HM, Raja A, Soos MP. *Physiology, Gastric Intrinsic Factor*. StatPearls Publishing; 2023. Updated July 17, 2023. Accessed June 2, 2025. https://www.ncbi.nlm.nih.gov/books/NBK546655/

Rapid Gastric Emptying:

- Goyal RK, Cristofaro V, Sullivan MP. Rapid gastric emptying in diabetes mellitus: pathophysiology and clinical importance. *J Diabetes Complications*. 2019;33(11):107414. doi:10.1016/j.jdiacomp.2019.107414
- Nguyen L, Wilson LA, Miriel L, et al. Autonomic function in gastroparesis and chronic unexplained nausea and vomiting: relationship with etiology, gastric emptying, and symptom severity. *Neurogastroenterol Motil*. 2020;32(8):e13810. doi:10.1111/nmo.13810

Gastroparesis:

- Reddivari AKR, Mehta P. *Gastroparesis*. StatPearls Publishing; 2024. Updated February 28, 2024. Accessed June 2, 2025. https://www.ncbi.nlm.nih.gov/books/NBK551528/

Eating Disorders:

- Keel PK, Eckel LA, Hildebrandt BA, et al. Disentangling the links between gastric emptying and binge eating v. purging in eating disorders using a case-control design. *Psychol Med*. 2023;53(5):1947-1954. doi:10.1017/S0033291721003640

Medications

Mediators of Digestive Enzyme Dysfunction

Agent	Mechanism
Proton pump inhibitors	Inhibits the H+/K+ ATPase enzyme in gastric parietal cells Reduces acid secretion, decreases stimulation of digestive enzyme release
Antacids	Neutralizes HCl, inhibiting pepsin
H2 receptor blockers	Blocks histamine receptors in stomach lining, reducing acid production
Somatostatin analogues	Somatostatin naturally inhibits the secretion of various digestive enzymes and hormones
Anticholinergics	Reduces the secretion of gastric acid and digestive enzymes by blocking acetylcholine's effects on GI track
Synthetic prostaglandins	Reduces gastric acid secretion while protecting the stomach lining
Prokinetics	Stimulates contractions and motility, reduces lower esophageal sphincter pressure

See: “References: Medications as Mediators of Digestive Dysfunction”

References

MEDICATIONS AS MEDIATORS OF DIGESTIVE DYSFUNCTION

- Yago MR, Frymoyer AR, Smelick GS, et al. Gastric reacidification with betaine HCl in healthy volunteers with rabeprazole-induced hypochlorhydria. *Mol Pharm*. 2013;10(11):4032-4037. doi:10.1021/mp4003738
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- Al-Frejat Z, Martini N, Esper A, Al-Frejat D, Younes S, Hanna M. GERD: latest update on acid-suppressant drugs. *Curr Res Pharmacol Drug Discov*. 2024;7:100198. doi:10.1016/j.crphar.2024.100198
- Laine L, DeVault K, Katz P, et al. Vonoprazan versus lansoprazole for healing and maintenance of healing of erosive esophagitis: a randomized trial. *Gastroenterology*. 2023;164(1):61-71. doi:10.1053/j.gastro.2022.09.041

Impairments in Digestion and Absorption

- Inadequate mastication and salivary secretions
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- Gastric dysmotility
- Pancreatic insufficiency
- Bile insufficiency
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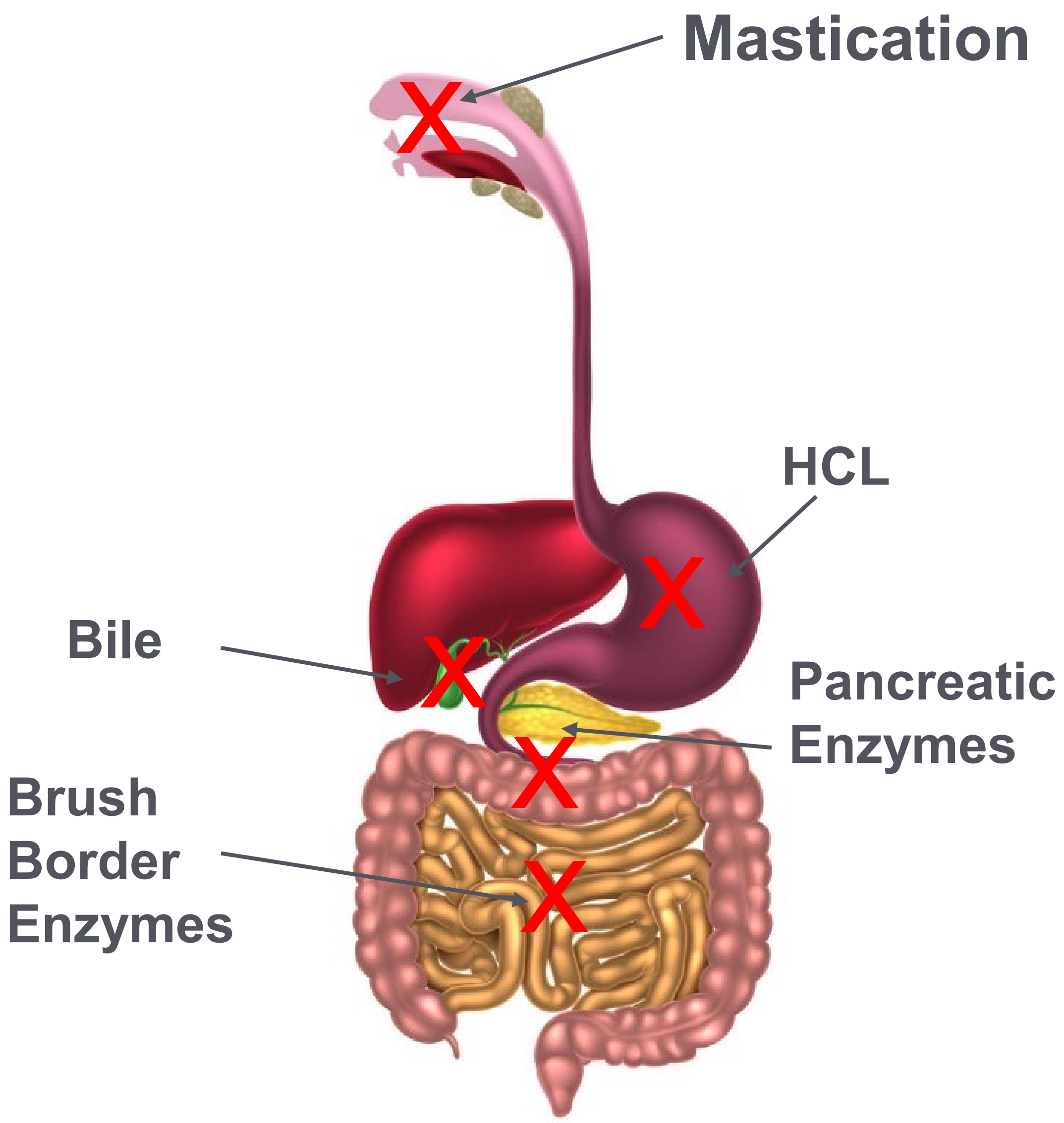

Mechanisms, Mediators, and Manifestations of Maldigestion/Malabsorption: *Pancreatic and Gallbladder Factors*

Mechanism	Substrate/Nutrients Affected	Antecedents, Triggers, Mediators
Bile acid insufficiency	**Fat, fat-soluble vitamins, calcium, magnesium**	Biliary obstruction, CCK deficiency, medications Liver disease, biliary disease
Bile acid malabsorption	**Fat, fat-soluble vitamins, calcium, magnesium**	Enteropathy, colitis, surgery (cholecystectomy), post-infectious diarrhea, SIBO, pancreatic insufficiency, pelvic radiation, medications (metformin) Ileal disease (Crohn’s disease)
Pancreatic insufficiency	**Fat, protein, carbohydrate, fat-soluble vitamins**	Genetic, congenital, pancreatic tumors, hyperacidity (inactivating pancreatic enzymes), biliary obstruction, surgery Cystic fibrosis, chronic pancreatitis, biliary disease, autoimmune conditions (Sjögren’s, primary biliary cirrhosis, IBD, celiac disease)

See: “References: Pancreatic and Gallbladder Factors”

References

PANCREATIC AND GALLBLADDER FACTORS

Bile Acid Insufficiency, Bile Acid Malabsorption, Pancreatic Insufficiency:

- Farrugia A, Arasaradnam R. Bile acid diarrhoea: pathophysiology, diagnosis and management. *Frontline Gastroenterol*. 2020;12(6):500-507. doi:10.1136/flgastro-2020-101436

Pancreatic Insufficiency:

- Ghodeif AO, Azer SA. *Pancreatic Insufficiency*. StatPearls Publishing; 2023. Updated January 16, 2023. Accessed June 2, 2025. https://www.ncbi.nlm.nih.gov/books/NBK555926/
- Omer A, Quigley EMM. Carbohydrate maldigestion and malabsorption. *Clin Gastroenterol Hepatol*. 2018;16(8):1197-1199. doi:10.1016/j.cgh.2018.01.048
- Owens SR, Greenson JK. The pathology of malabsorption: current concepts. *Histopathology*. 2007;50(1):64-82. doi:10.1111/j.1365-2559.2006.02547

Medications:
Mediators of Pancreatic and Bile Insufficiency

Agent	Mechanism
Proton pump inhibitors	Reduce pancreatic enzyme secretion
Antibiotics	Disrupt gut microbiota and potentially damage pancreatin tissue, leading to EPI; can cause cholestatic liver injury by damaging bile ducts and inhibiting bile flow
Steroids	Can cause pancreatic ductal cell dysfunction, leading to pancreatitis
Diuretics	Can induce hypercalcemia
Immunosuppressants	Can cause direct pancreatic toxicity
Estrogen (contraceptives or HRT)	Can impair bile secretion
NSAIDs	Effects liver cells and bile ducts, leading to cholestasis
Psychotropic medications	May induce cholestatic hepatitis through immune-mediated bile duct injury

See: “References: Medications as Mediators of Pancreatic and Bile Insufficiency”

References

MEDICATIONS AS MEDIATORS OF PANCREATIC AND BILE INSUFFICIENCY

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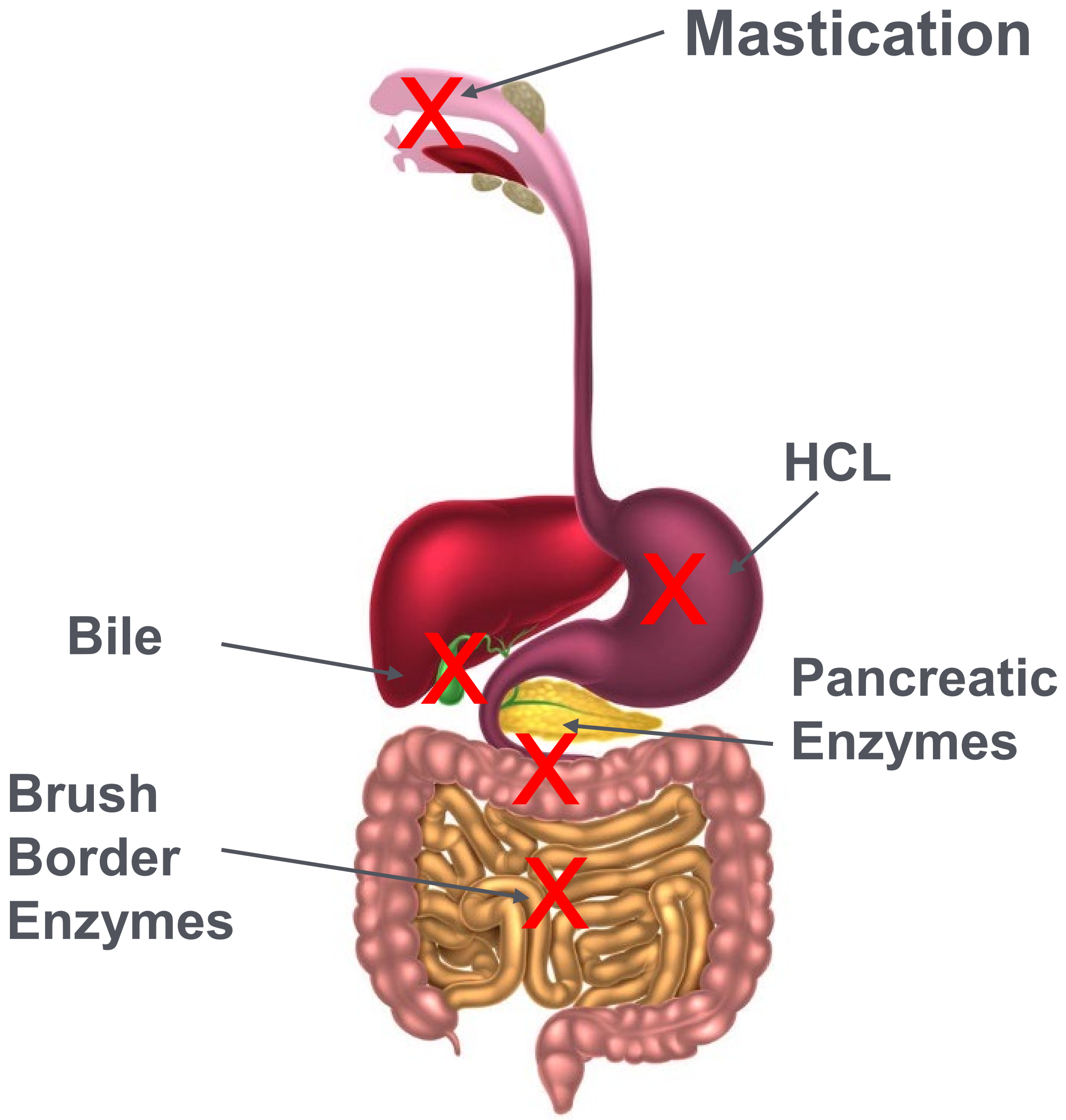
Impairments in Digestion and Absorption

- Inadequate mastication and salivary secretions
- Hypochlorhydria
- Gastric dysmotility
- Pancreatic insufficiency
- Bile insufficiency
- Brush border injury
- Intestinal dysmotility



Mechanisms, Mediators, and Manifestations of Maldigestion/Malabsorption: *Brush Border Factors*

Mechanism	Substrate/Nutrients Affected	Antecedents, Triggers, Mediators
Impaired mucosal digestion	**CHO, protein**	Brush border enzyme deficiency (i.e., lactase, sucrase) Mucosal disease (i.e., Crohn's, celiac)
Reduced mucosal absorption	**Fat, CHO, protein, vitamins, minerals**	Intestinal surgery, infections, malignancy Mucosal disease (i.e., Crohn's, celiac)

See: "References: Brush Border Factors"

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BRUSH BORDER FACTORS

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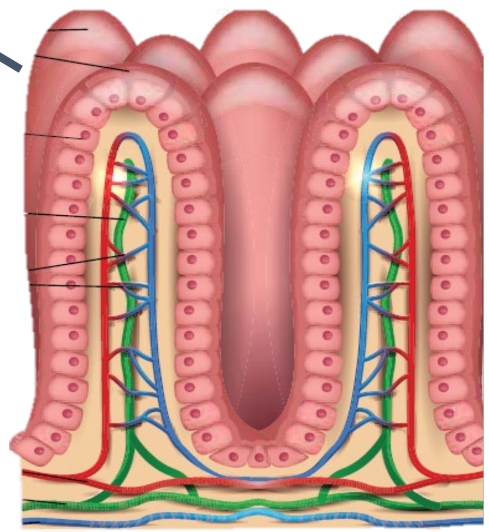
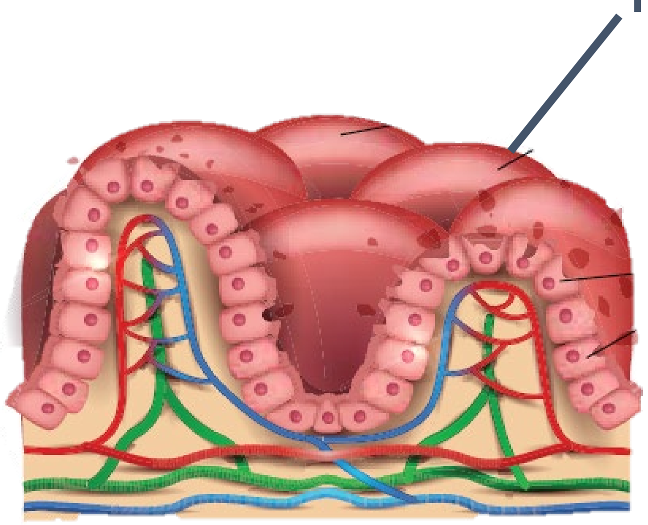
Brush Border Injury Example

Celiac Disease

Flattened microvilli in celiac disease





Medications as Mediators of Brush Border Injury

	Maldigestion	Malabsorption
NSAIDs	X	X
Antibiotics	X	
Proton pump inhibitors	X	
Olmesartan and ACE inhibitors		X
Mycophenolate mofetil		X
Chemotherapeutics		X
Immune checkpoint inhibitors		X
Contraceptives	X	X
Anticoagulants	X	X

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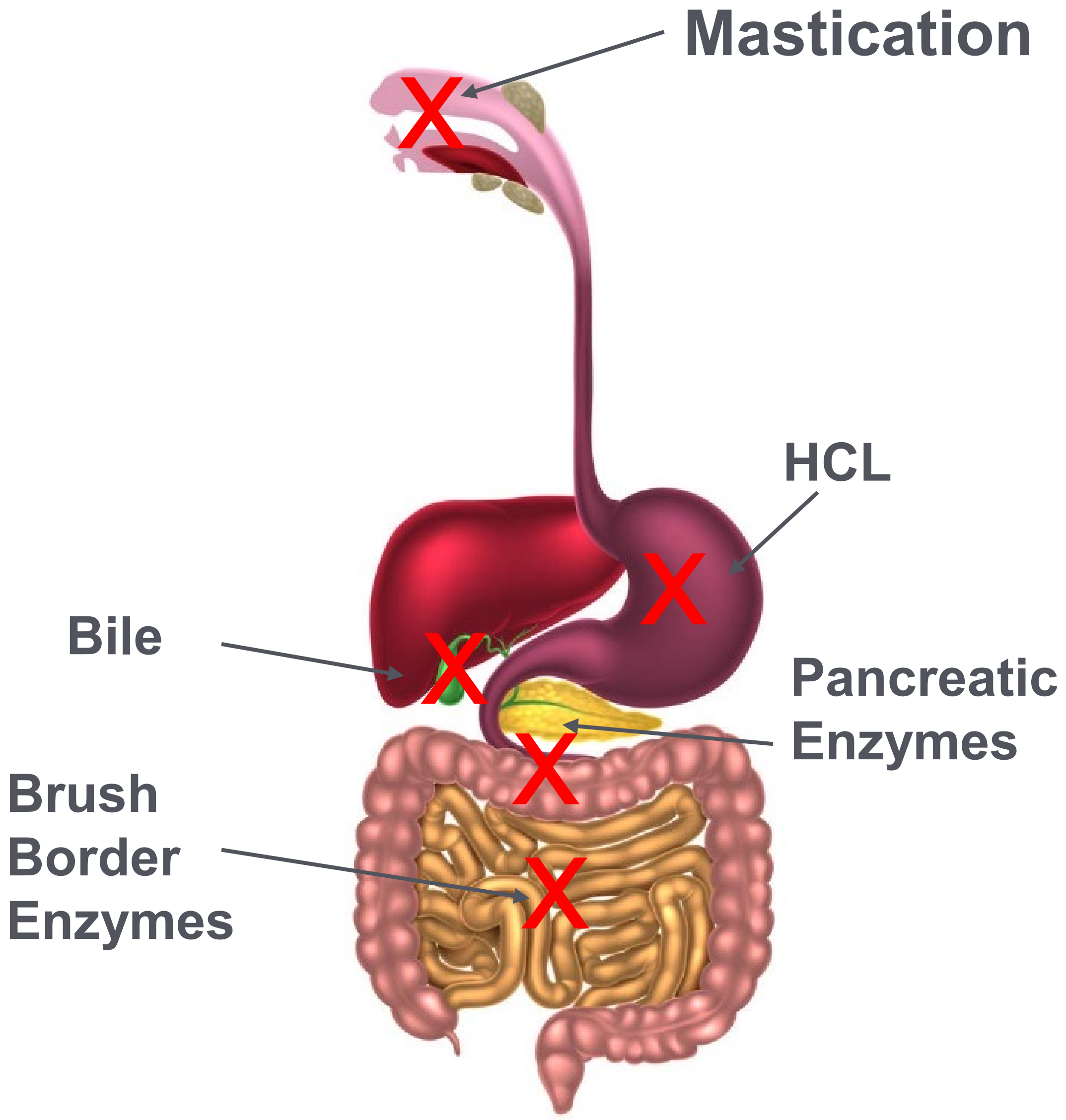
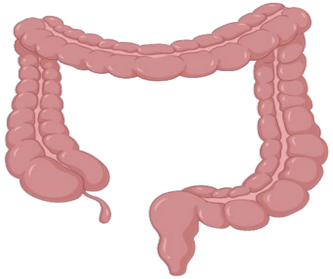
Impairments in Digestion and Absorption

- Inadequate mastication and salivary secretions
- Hypochlorhydria
- Gastric dysmotility
- Pancreatic insufficiency
- Bile insufficiency
- Brush border injury
- Intestinal dysmotility

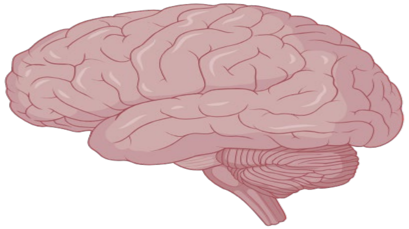


Intestinal Phase of Digestion: Motility



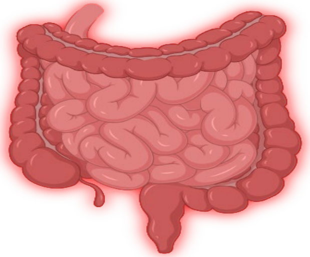
Function

- Mix undigested food with enzymes and secretions
- Move content at a constant rate for optimal digestion and absorption



ENS Mediation

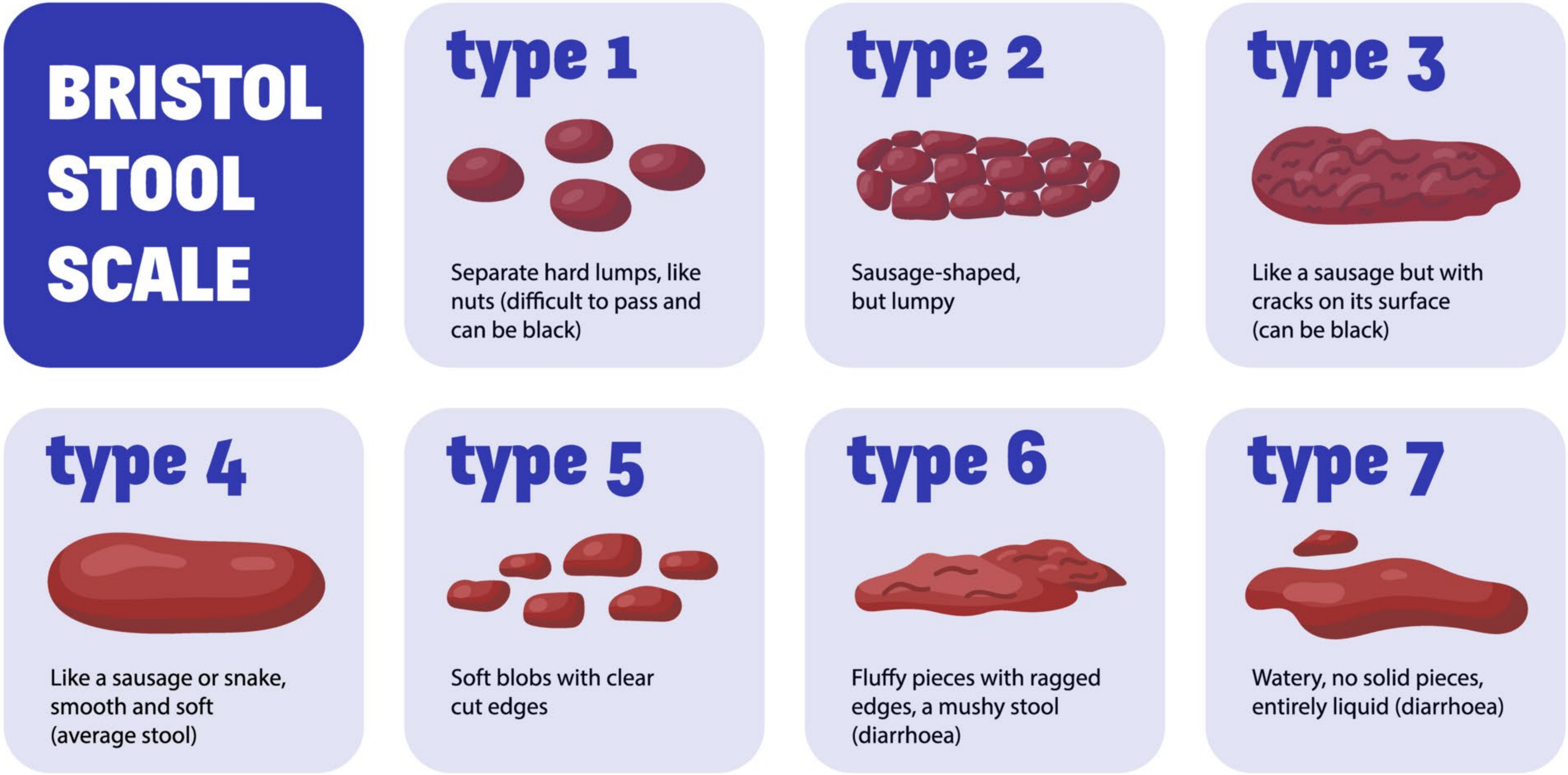
- Peristaltic waves occur frequently in the small intestine and 2-4 times daily in the large intestine (most active following a meal)



Motility Disorders

GERD, SIBO, gastroparesis, achalasia, congenital disorders (Hirschsprung's disease), age-related atrophy

Images: Created with BioRender.com by The Institute for Functional Medicine


BRISTOL STOOL SCALE
type 1
Separate hard lumps, like nuts (difficult to pass and can be black)
type 2
Sausage-shaped, but lumpy
type 3
Like a sausage but with cracks on its surface (can be black)
type 4
Like a sausage or snake, smooth and soft (average stool)
type 5
Soft blobs with clear cut edges
type 6
Fluffy pieces with ragged edges, a mushy stool (diarrhoea)
type 7
Watery, no solid pieces, entirely liquid (diarrhoea)

Long Description: Bristol Stool Scale

Bristol Stool Scale

- Type 1: Separate hard lumps, like nuts (difficult to pass and can be black)
- Type 2: Sausage-shaped but lumpy
- Type 3: Like a sausage but with cracks on the surface (can be black)
- Type 4: Like a sausage or snake, smooth and soft (average stool)
- Type 5: Soft blobs with clear cut edges
- Type 6: Fluffy pieces with ragged edges, a mushy stool (diarrhea)
- Type 7: Watery, no solid pieces, entirely liquid (diarrhea)

Mechanisms, Mediators, and Manifestations of Maldigestion/Malabsorption: *Motility and Microbial Factors*

Mechanism	Substrate/Nutrients Affected	Antecedents, Triggers, Mediators
Intraluminal consumption of nutrients (microbial causes)	**B12, macronutrients**	Parasitic infection (i.e., *D. latum*, *Giardiasis*, Cryptosporidiosis, *Microsporidiosis*) SIBO
Dysmotility (hypomotility, hypermotility)	**Fat, protein**	Lymphatic disease, venous stasis (i.e., CHF)

See: "References: Motility and Microbial Factors"

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MOTILITY AND MICROBIAL FACTORS

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Medications
Mediators of Dysmotility

Agent	Mechanism
Opioids (C)	Reduce peristalsis and increase fluid absorption
Anti-diarrheal agents (C)	Inhibit peristalsis, slowing transit
Anticholinergic drugs (C)	Reduce smooth muscle contractions
Ca^{2+} channel blockers (C)	Relax smooth muscle, leading to decreased motility and constipation
GLP-1 receptor agonists (C)	Can lead to delayed gastric emptying, reduce small intestinal motility, suppress parasympathetic outflows
Prokinetics (D)	Antagonize dopamine receptors and increase release of acetylcholine to stimulate peristalsis
Macrolide antibiotics (D)	Can mimic motilin, a hormone that stimulates gastric and small intestinal motility, aiding in conditions like gastroparesis
Serotonin agonists (D)	Stimulate 5HT receptors to enhance peristalsis and accelerate colonic transit

See: “References: Medications as Mediators of Dysmotility”

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MEDICATIONS AS MEDIATORS OF DYSMOTILITY

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ATMs of Maldigestion & Malabsorption

The Patient's Story

Antecedents (Inherited & Acquired)

- Family history of celiac disease

Triggers & Precipitating Events

- Cholecystectomy

Mediators/Perpetuators

- Chronic NSAID use
- GLP-1

Antecedents, Triggers, Mediators

Modifiable Lifestyle Factors

Modifiable Lifestyle Factors

Sleep

- Inadequate sleep or disrupted sleep cycle

Physical Activity

- Sedentarism
- Overexercise

Nutrition

- Ultra-processed diet
- Dehydration

Stressors

- Life stressors

Relationships

- Loneliness
- Isolation

See: "References: ATMs of Digestion"

Long Description: ATMs of Maldigestion and Malabsorption

- Antecedents
  - Family history of Celiac disease
- Triggers
  - Cholecystectomy
- Mediators
  - Chronic N SAID use
  - G L P 1
- Modifiable lifestyle factors
  - Sleep: Inadequate sleep or disrupted sleep cycle
  - Physical activity: sedentarism, overexercise
  - Nutrition: Ultra-processed diet, dehydration
  - Stressors: Life stressors
  - Relationships: Loneliness, isolation

References

ATMS OF DIGESTION

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Clinical Focus

- Clinical findings of maldigestion and malabsorption.

The ABCDs of Nutrition Evaluation

Representative Clinical Findings

Hypochlorhydria & Gastric Dysmotility

Early Satiety

Dyspepsia <1 hr Postprandial

Constipation

Multiple Food Allergies

Undigested Food in Stool

Weak/Brittle Peeling Fingernails

See: “References: Most Common Clinical Indicators of Maldigestion & Malabsorption”
Image: Created with BioRender.com by The Institute for Functional Medicine

Clinical Findings
Hypochlorhydria & Gastric Dysmotility

Image: Landkartenzunge by Bin im Garten

Clinical Findings

Hypochlorhydria & Gastric Dysmotility

Gastrointestinal Signs and Symptoms

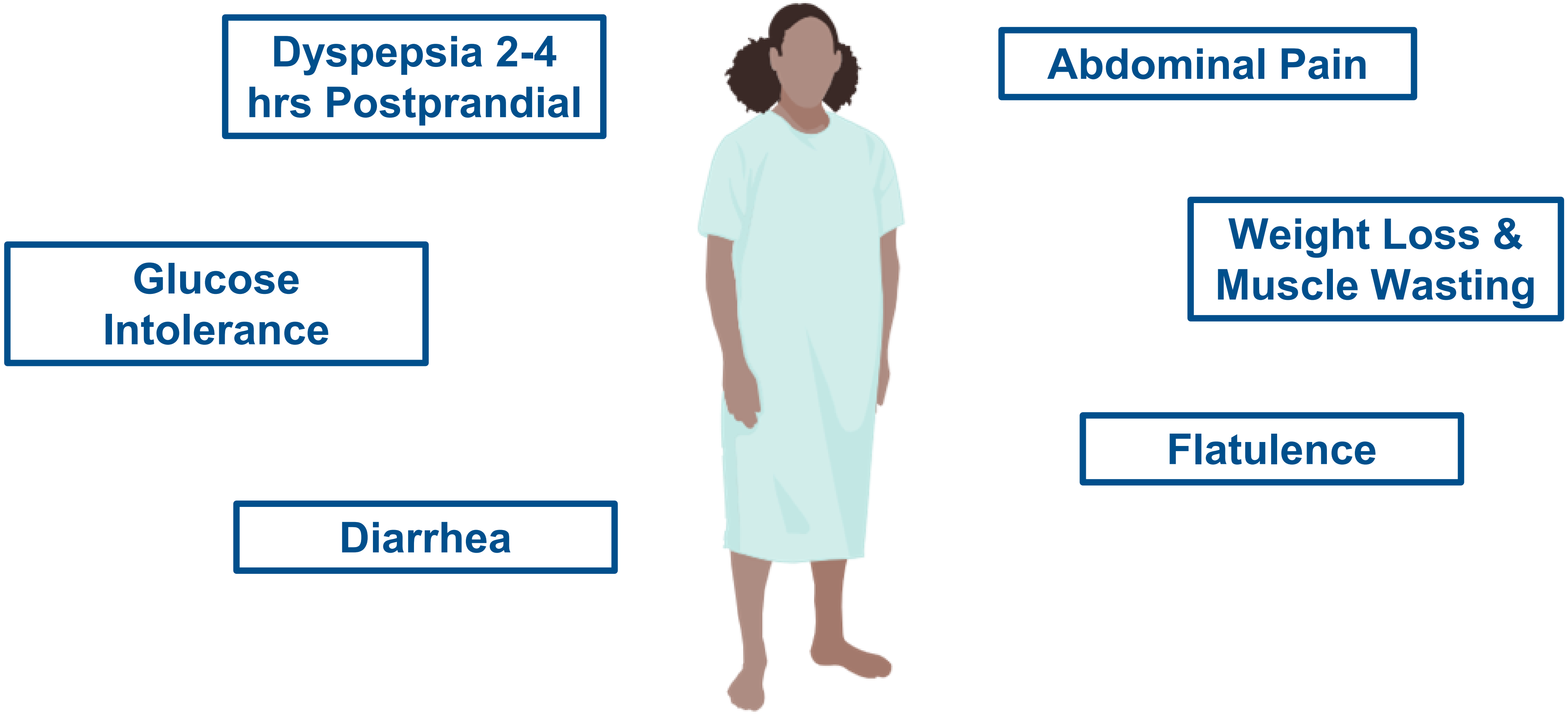
- Bloating or belching (postprandial)
- A sense of fullness after eating
- Early satiety
- Epigastric pain or abdominal pain
- Diarrhea/constipation
- Acid regurgitation
- Vomiting
- Undigested food in stool
- Chronic intestinal infections

Extraintestinal Signs and Symptoms

- Weak, peeling, or cracked fingernails
- Glossitis
- Post-adolescent acne
- Dilated capillaries on the face (acne rosacea)
- Nutrient deficiencies (iron, B12, vitamin D, calcium, zinc)
- Multiple food allergies and sensitivities

See: “References: Most Common Clinical Indicators of Maldigestion & Malabsorption”

Representative Clinical Findings
Pancreatic Exocrine Dysfunction



See: “References: Most Common Clinical Indicators of Maldigestion & Malabsorption”
Image: Created with BioRender.com by The Institute for Functional Medicine

Clinical Findings
Pancreatic Exocrine Dysfunction

Clinical Findings
Pancreatic Exocrine Dysfunction

Gastrointestinal Signs and Symptoms
- Steatorrhea
- Flatulence
- Abdominal pain
- Indigestion/fullness 2-4 hours after a meal
- Bloating or flatulence 2-4 hours after a meal
- Undigested food in stool
- Bloating
- Diarrhea
- Abnormal stool frequency
- Bulky stool

Extraintestinal Signs and Symptoms
- Weight loss
- Muscle wasting
- Fatigue
- Glucose intolerance
- Vitamin deficiencies
- Edema (hypoalbuminemia)

See: “References: Most Common Clinical Indicators of Maldigestion & Malabsorption”

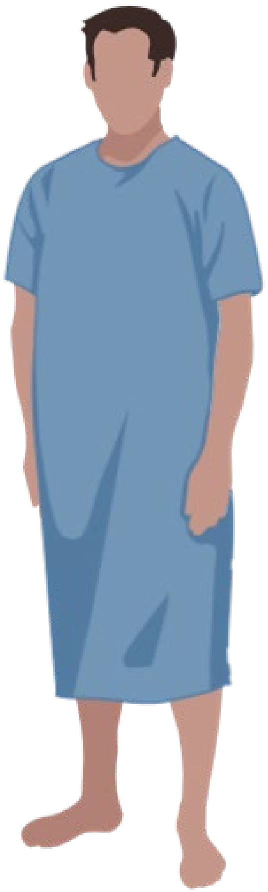
Representative Clinical Findings

Bile Acid Insufficiency

Cholestasis

Fatigue

Pruritus



Poor Night Vision

Swelling

Steatorrhea

See: “References: Most Common Clinical Indicators of Maldigestion & Malabsorption”
Image: Created with BioRender.com by The Institute for Functional Medicine

Clinical Findings
Bile Acid Insufficiency

Image courtesy of Michael Stone, MD

Clinical Findings
Bile Acid Insufficiency

Gastrointestinal Signs and Symptoms

- Steatorrhea
- Epigastric or right upper quadrant pain (cholestasis)
- Nausea

Extraintestinal Signs and Symptoms

- Pruritus (cholestasis)
- Xanthomas (cholestasis)
- Weight loss
- Nutritional deficiencies (fat-soluble vitamins)
- Fatigue
- Signs/symptoms of liver disease
- Signs/symptoms of biliary conditions

See: "References: Most Common Clinical Indicators of Maldigestion & Malabsorption"

Representative Clinical Findings
Brush Border Injury

Dyspepsia >1 hr Postprandial

Muscle/Joint Pain

Osteoporosis

Abdominal Pain

Headache

Non-Celiac Gluten Sensitivity

Diarrhea

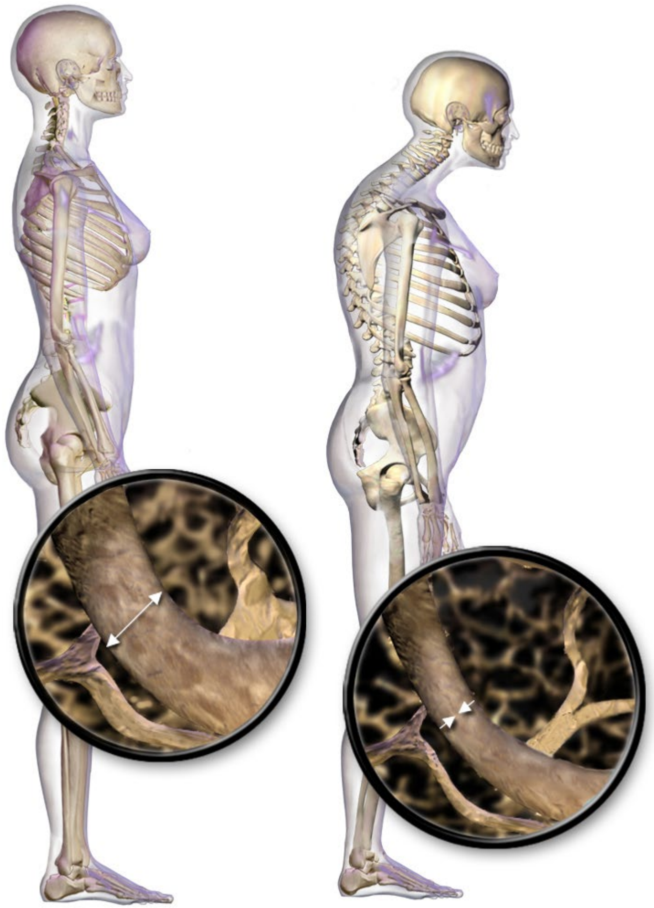
Skin Lesions

Flatulence

See: “References: Most Common Clinical Indicators of Maldigestion & Malabsorption”
Image: Created with BioRender.com by The Institute for Functional Medicine

Clinical Findings

Brush Border Injury



Clinical Findings

Brush Border Injury

Gastrointestinal Signs and Symptoms

Post-prandial symptoms of:

- Abdominal pain
- Gas
- Bloating
- Diarrhea
- Non-celiac gluten sensitivity

Extraintestinal Signs and Symptoms

Associated with lactose intolerance:

- Nutritional deficiencies
- Fatigue
- Headache
- Cognitive issues
- Muscle/joint pain
- Skin lesions
- Mouth ulcers
- Heart palpitations
- Osteoporosis

See: "References: Most Common Clinical Indicators of Maldigestion & Malabsorption"

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Maldigestion and Malabsorption:
Nutrient-Specific Clinical Findings

Macro/Micronutrients	Clinical Indicators
Fat	Pale, voluminous, greasy offensive diarrhea
Protein	Edema, muscle atrophy, amenorrhea
Carbohydrate	Abdominal bloating, flatus, diarrhea
B12	Macrocytic anemia, SSCD spinal cord
Folate	Macrocytic anemia
B Vitamins	Cheilosis, glossitis, stomatitis, acrodermatitis
Iron	Microcytic anemia
Calcium, Vitamin D	Osteomalacia, bone pain, fractures, tetany
Vitamin A	Follicular hyperkeratosis, night blindness
Vitamin K	Bleeding diathesis, hematoma

See: “References: Macronutrients, Micronutrients, and Malabsorption”

References

MACRONUTRIENTS, MICRONUTRIENTS, AND MALABSORPTION

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Clinical Focus

- Summary, conclusion, and key clinical takeaways.

Strive for Harmony

Phases of Digestion

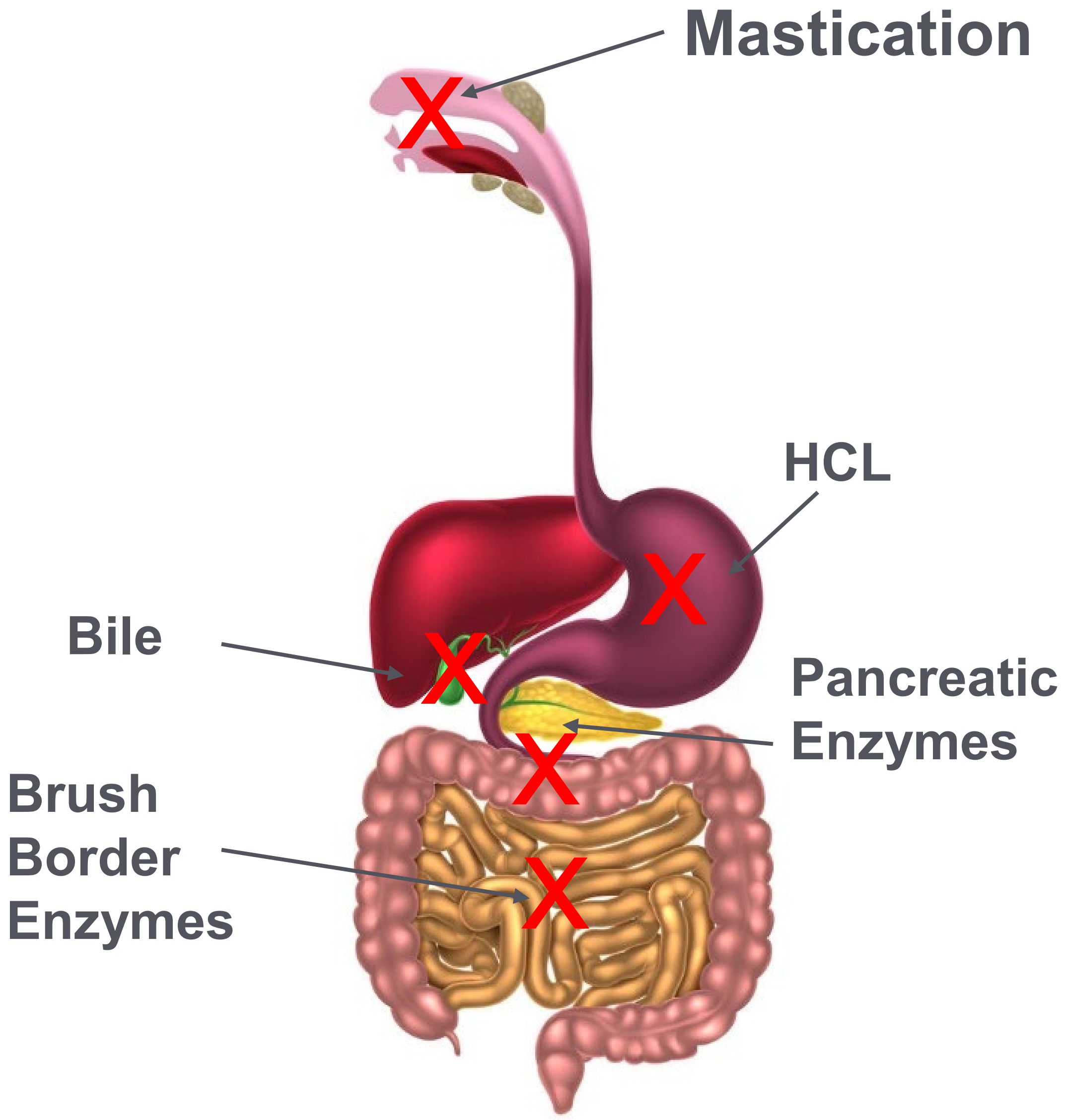
Cephalic phase

Gastric phase

Intestinal phase

Impairments in Digestion and Absorption

- Inadequate mastication and salivary secretions
- Hypochlorhydria
- Gastric dysmotility
- Pancreatic insufficiency
- Bile insufficiency
- Brush border injury
- Intestinal dysmotility



Maldigestion and Malabsorption:

Manifestations – Mediators – Mitigators

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE